Topic 1: Mathematical Expectation

1. What is expectation?

Definition: Expected value is the weighted average of all possible values of a random variable, where probabilities act as weights.

In simple words: A random variable can take different values. Expectation tells us its average value over many repeated experiments.

2. Formula for a discrete random variable

$$E[X] = \sum_{x} xP(X = x)$$

Meaning:

- x = possible value of random variable.
- $P(X = x)$ = probability of that value.
- Multiply each value by its probability and add everything.

3. Solved example – Fair die

A fair die is rolled once. Let X be the number appearing.

Possible values:

$$X = \{1, 2, 3, 4, 5, 6\}$$

Each value has probability $1/6$.

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \cdots + 6\left(\frac{1}{6}\right)$$

$$= \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \boxed{3.5}$$

Topic 2: Properties of Expectation

These properties are very important for GATE DA.

Property 1: Expectation of a constant

$$E[c] = c$$

Example:

$$E[5] = 5$$

Property 2: Constant multiplication

$$E[aX] = aE[X]$$

Example: If $E[X] = 4$, then

$$E[3X] = 3(4) = 12$$

Property 3: Addition of a constant

$$E[X + c] = E[X] + c$$

Example: If $E[X] = 4$,

Property 4: Linearity of expectation

$$E[X + Y] = E[X] + E[Y]$$

Important GATE point: **This property does not require X and Y to be independent.**

Expectation mini test

Q1. A fair coin is tossed once. X = 1 for Head and 0 for Tail. Find E[X].

- A. 0
- B. 1/4
- C. 1/2
- D. 1

Q2. If E[X] = 5, find E[3X + 2].

- A. 15
- B. 17
- C. 10
- D. 7

Q3. For any two random variables X and Y, which statement is always true?

- A. E[X+Y] = E[X]E[Y]
- B. E[X+Y] = E[X]+E[Y]
- C. X and Y must be independent
- D. E[XY] = E[X]E[Y]

Topic 3: Variance of a Random Variable

1. What is variance?

Variance measures how far the possible values of a random variable are spread around their expected value.

2. Formula

The most important GATE DA formula:

$$\boxed{Var(X) = E[(X - \mu)^2]}$$

Where:

- X = Random variable
- $\mu = E[X]$ = Mean or expected value
- $X - \mu$ = Deviation from mean

An equivalent and very useful formula:

$$\boxed{Var(X) = E[X^2] - (E[X])^2}$$

GATE shortcut Remember this second formula. It is often faster for numerical problems.

3. Solved example

Suppose a random variable X has the following distribution:

X	P(X)
1	0.5
3	0.5

Step 1: Calculate expectation

$$E[X] = 1(0.5) + 3(0.5)$$

$$= 0.5 + 1.5 = 2$$

Step 2: Calculate $E[X^2]$

$$E[X^2] = 1^2(0.5) + 3^2(0.5)$$

$$= 0.5 + 4.5 = 5$$

Step 3: Calculate variance

$$Var(X) = E[X^2] - (E[X])^2$$

$$= 5 - (2)^2$$

$$\boxed{Var(X) = 1}$$

4. Important properties of variance

Property 1: Variance of a constant

$$Var(c) = 0$$

Property 2: Adding a constant

$$Var(X + c) = Var(X)$$

Adding a constant shifts all values but doesn't change their spread.

Property 3: Multiplying by a constant

$$Var(aX) = a^2 Var(X)$$

Property 4: General formula

$$\boxed{Var(aX + b) = a^2 Var(X)}$$

Remember: The constant b disappears, but a gets squared.

5. Standard deviation

Standard deviation is the square root of variance.

$$\boxed{\sigma = \sqrt{Var(X)}}$$

$$\sigma = \sqrt{1} = 1$$

Variance mini test

Q1. If E[X] = 4 and $E[X^2] = 20$, find Var(X).

- A. 4
- B. 16
- C. 20
- D. 36

Q2. If Var(X) = 5, find Var(3X + 7).

- A. 15
- B. 22
- C. 45
- D. 52

Q3. If Var(X) = 8, what is Var(X + 100)?

- A. 108
- B. 100
- C. 8
- D. 0

Q4. If X is a constant random variable equal to 7, its variance is:

- A. 7
- B. 49
- C. 1
- D. 0

Variance formula revision sheet

Rule	Formula
Basic variance	$Var(X) = E[X^2] - (E[X])^2$
Constant	$Var(c) = 0$
Adding constant	$Var(X + b) = Var(X)$
Multiplying constant	$Var(aX) = a^2 Var(X)$
Combined	$Var(aX + b) = a^2 Var(X)$

Topic 4: Expectation of a function of X

Sometimes the question gives X, but asks us to calculate X^2, X^3, or another function of X.

The formula is:

$$\boxed{E[g(X)] = \sum_x g(x)P(X = x)}$$

Example

Suppose:

X	P(X)
1	0.5
2	0.5

Find $E[X^2]$.

$$E[X^2] = 1^2(0.5) + 2^2(0.5)$$

$$= 0.5 + 2 = \boxed{2.5}$$

Important difference:

$$E[X^2] \neq (E[X])^2$$

Let's verify:

$$E[X] = 1(0.5) + 2(0.5) = 1.5$$

$$(E[X])^2 = 1.5^2 = 2.25$$

But $E[X^2] = 2.5$.

This difference is exactly why variance can be calculated as:

$$Var(X) = E[X^2] - (E[X])^2$$

Topic 5: Continuous random variable – Expectation and variance

For a continuous random variable with PDF $f(x)$:

Expected value:

$$\boxed{E[X] = \int_{-\infty}^{\infty} x f(x)\, dx}$$

Expected square:

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx$$

Variance:

$$\boxed{Var(X) = E[X^2] - (E[X])^2}$$

Solved example

Let the PDF be:

$$f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Find expectation.

$$E[X] = \int_0^1 x(1)\,dx$$

$$= \left[\frac{x^2}{2}\right]_0^1 = \boxed{\frac{1}{2}}$$

Find $E[X^2]$:

$$E[X^2] = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

Find variance:

$$Var(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{3} - \frac{1}{4} = \boxed{\frac{1}{12}}$$

$$\boxed{Var(X) = \frac{1}{12}}$$

Topic 6: Important formula revision

Concept	Discrete	Continuous
Expectation	$\sum xf(x)$	$\int xf(x)dx$
Expected square	$\sum x^2p(x)$	$\int x^2f(x)dx$
Variance	$E[X^2] - E[X]^2$	$E[X^2] - E[X]^2$

Mixed practice test

Q1. If E[X] = 3 and $E[X^2]$ = 11, find Var(X).

- A. 2
- B. 9
- C. 11
- D. 20

Q2. If E[X] = 4, what is E[2X + 5]?

- A. 8
- B. 9
- C. 13
- D. 18

Q3. For a continuous random variable, $E[X^2]$ is calculated using:

- A. ∫ x f(x) dx
- B. ∫ x^2 f(x) dx
- C. Σ x^2
- D. ∫ f(x) dx

Q4. If E[X] = 2 and Var(X) = 5, find $E[X^2]$.

- A. 1
- B. 5
- C. 7
- D. 9

Q5. For the uniform PDF f(x)=1 on $0 \le x \le 1$, find E[X].

- A. 0
- B. 1/2
- C. 1/3
- D. 1

Section A – MCQs

Q1. A random variable X has the following distribution:

X	0	1	2
P(X)	0.2	0.5	0.3

Find $E[X]$.

A. 0.8

B. 1.1

C. 1.3

D. 1.5

Q2. If $E[X] = 4$ and $Var(X) = 9$, find $E[X^2]$.

A. 13

B. 16

C. 25

D. 36

Q3. If $Var(X) = 5$, find $Var(3X - 7)$.

A. 15

B. 38

C. 45

D. 52

Q4. A fair coin is tossed twice. Let X denote the number of heads. Find $Var(X)$.

A. 0.25

B. 0.5

C. 1

D. 2

Q5. A continuous random variable has PDF:

$$f(x) = 2x, \quad 0 \leq x \leq 1$$

Find $E[X]$.

A. $1/3$

B. $1/2$

C. $2/3$

D. 1

Section B – MSQs

Select all correct options.

Q6. Which statements about expectation are always true?

A. $E[X+Y] = E[X] + E[Y]$

B. $E[aX] = aE[X]$

C. $E[XY] = E[X]E[Y]$ for all random variables

D. $E[c] = c$

Q7. Which statements about variance are correct?

A. $Var(X+c) = Var(X)$

B. $Var(aX) = aVar(X)$

C. $Var(aX+b) = a^2 Var(X)$

D. $Var(X) \geq 0$

Q8. Let X be a random variable with finite second moment. Which statements are correct?

A. $Var(X) = E[X^2] - (E[X])^2$

B. $E[X^2] \geq (E[X])^2$

C. $Var(X) = 0$ implies X is constant with probability 1

D. $E[X^2] = (E[X])^2$ always

Section C – NAT

Enter numerical answers.

Q9. A random variable X has:

$$E[X] = 5, \quad Var(X) = 4$$

Find:

$$E[(X-2)^2]$$

Q10. A continuous random variable has PDF:

$$f(x) = \begin{cases} 3x^2, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Calculate its variance.

Give your answer up to 3 decimal places.